

HexaRescue – Hexapod mountain rescue robot

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Abstract — *This article presents the early development of HexaRescue, a hexapod [1] mountain rescue robot. The project focuses on creating a 3-DOF [2] leg architecture actuated by servomotors. Each prototype was individually tested. At this stage, three legs have been assembled and were set in motion. These early results confirm the feasibility of this hexapod but also the reliability of the chosen design.*

Index terms — *Hexapod, legged robot, Arduino, servomotor, inverse kinematics*

I. INTRODUCTION

Hexapod robots offer advantages compared to wheeled robots. They can handle obstacles better and have greater stability on uneven terrain. HexaRescue is based on this idea. It is a 6-legged robot designed to assist rescue teams and reduce intervention time. Each leg has 3 degree of freedom (DOF). They are controlled by an SSC-32 board [3] and Arduino UNO [4]. The main structure is in metal, and the legs are 3D-printed in Polyethylene Terephthalate Glycol (PETG). The main goals of HexaRescue are to move on irregular terrain, detect victims using a thermal camera, and transmit their GPS position to rescue teams.

This article is organized as follows. Section II presents the mechanical design of the robot. Section III describes the electronic architecture and the components used. Section IV introduces the control code and the first locomotion tests. Finally, Section V presents the conclusions.

II. MECHANICAL DESIGN

The first version of the robot was made entirely in metal. It was a kit with servomotors and metal legs. However, this first version had several limitations. Firstly, the legs were too short so the robot would have difficulties straddling branches or rocks. Secondly, the build quality was insufficient. So, it was decided to keep only the metal body and 3D print the legs. This compromise gives HexaRescue good structural strength but also a strong and durable base for mounting the new legs.

The new legs were entirely designed using Computer-Aided Design (CAD) (Fig.1) with fusion360 [5]. Multiple versions were needed to obtain a leg that was rigid, practical, and aesthetically clean. Then, these legs were manufactured using a 3D printer and Computer-Aided Manufacturing (CAM). PETG is the filament used because it has good mechanical properties: lightweight, mechanical strength, and good shock resistance. Moreover, using CAD/CAM [6] and 3D printed legs is good for adjusting the model and allows modular parts that can easily be replaced.

Finally, a 3 DOF leg architecture was chosen for the legs because, compared to a 2 DOF leg architecture, it offers a wider range of motion. Thanks to a 3 DOF leg architecture, a leg can step over an obstacle by lifting its hip and rotating the leg. Some motions would not be possible without the 3rd DOF.

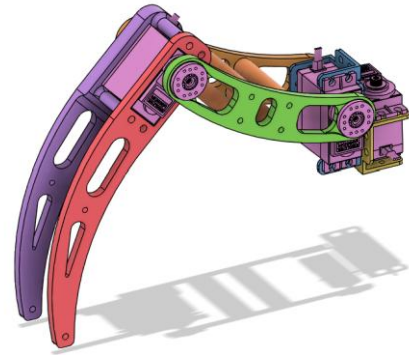


Figure 1 : CAD model of a single 3-DOF leg created in Fusion 360.

III. ELECTRONIC ARCHITECTURE

The electronic architecture of HexaRescue is composed of boards and sensors, as illustrated in Fig. 2.

A. Main electronics

The electronic architecture of HexaRescue is based on an Arduino UNO combined with an SSC-32 board, used because it can pilot several servomotors at the same time. This board is precise and reliable. The legs use servomotors TD-8135MG [7] with a force of 35 kg.cm. It offers high torque which is necessary to hold the weight of the robot, to facilitate motions, but also to carry some light tools. HexaRescue is now powered by a laboratory power supply, so it currently operates exclusively via wired power.

B. Sensors

The robot also integrates several sensors designed to improve its perception of the environment. A HC SR-04 [8], an ultrasonic sensor, is placed at the front of the body to detect obstacles and avoid front collisions. A thermal camera AMG8833 [9] is also fixed at the front of the robot. It can distinguish heat sources and is a key element for detecting victims in mountain environments. This combination of sensors provides an initial perception layer, which will be expanded as the prototype continues to develop.

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Finally, HexaRescue uses an HC-05 [10] Bluetooth module to be controlled by an application on a smartphone.

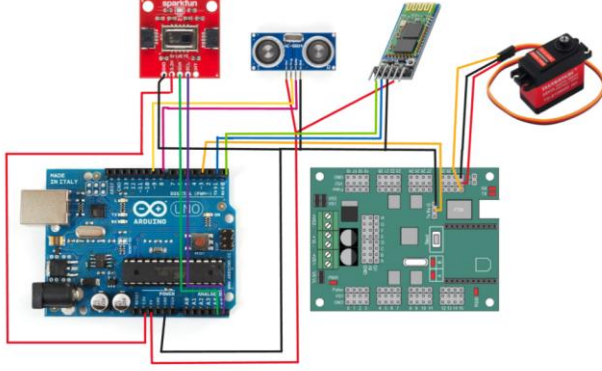


Figure 2 : Electronic achitecture of HexaRescue.

IV. LOCOMOTION AND CODE

A. Inverse kinematics and control of the legs

The locomotion of HexaRescue is based on the calculations of inverse kinematics equations [11]. This method allows converting a target position in space (x, y, z) into the corresponding angles for each servomotor ($\theta_1, \theta_2, \theta_3$), where θ_1, θ_2 , and θ_3 represent the hip yaw, hip pitch, and knee pitch angles respectively.

These calculations are the basis of every leg movement and that ensures that each foot follows the good trajectory with precision.

B. Locomotion modes and tripod gait

HexaRescue integrates different operational modes for the moment.

- Shut down mode: the robot is inactive. All the legs are pointing toward the sky, and the body is resting on the ground, as presented in fig.3.
- Enable mode: the legs move from their shut down position to the neutral position, and they slowly lift the robot into a stable posture, as presented in fig.4.
- Neutral position mode: each leg is in a stable position without any movement.
- Forward mode: the robot walks using a tripod gait, a gait in which three legs are moving while the others are stabilizing the robot, as shown in Table I.

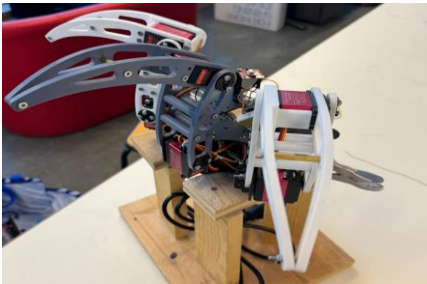


Figure 3 : Shut down mode.

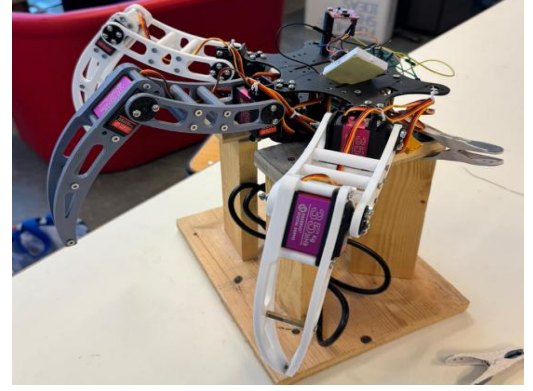


Figure 4 : Enable mode.

TABLE I. Table representing the tripod gait, with black cells indicating when the legs are in contact with the ground.

LEG 1					
LEG 2					
LEG 3					
LEG 4					
LEG 5					
LEG 6					

The height of the robot can also be adjusted dynamically. This allows the body of the robot to be raised or lowered depending on the user's need or the terrain.

C. Code structure

All the elements described above were integrated into the main program of HexaRescue. To improve accessibility and maintenance, the code was placed into different classes and files.

Each class has a precise role: one manages the legs, another handles the inverse kinematics calculations, and another communicates with the SSC-32 board to send all the data at the same time to the servomotors.

This organization helps keep the code readable and structured, and makes it easier to improve, such as new functions or sensors. All this code was written in the Arduino IDE [12].

V. CONCLUSION

HexaRescue is a hexapod robot designed to move in a stable way thanks to the inverse kinematics equations and the tripod gait which is simple but effective. At this stage, three legs have been assembled and tested but all the different functioning modes were correctly integrated and tested, as well as the sensors. The structure of the code is helpful to maintain the robot and to add new gaits or sensors.

It can be a lot more evolutive than before. This works is a solid base for HexaRescue, and to add new functionalities.

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